

Predictive Model for Heart Disease Based on Demographic, Societal, and Lifestyle Factors

Wilson Luong, David Mcglasson, Brayan Mejia Nunez,
Alejandro Oropeza, Hunter Powell, Moises Robledo
*Department of Computer Science, College of Engineering and Computer Science,
California State University, Sacramento, U.S.A*

Keywords: Cardiovascular Disease Prediction, Machine Learning, LightGBM, Demography, Lifestyle, Hyperparameter Tuning, Heart Disease.

Abstract: Machine learning for clinical decision support is rapidly expanding. While recent models successfully predict cardiovascular risk, they frequently rely on data-altering techniques like undersampling or synthetic generation to handle inherent class imbalances. This study develops an improved predictive model for heart-related conditions using the 2020 BRFSS dataset (319,795 respondents), leveraging demographic, societal, and lifestyle factors. A central challenge is inferring disease risk under severe class imbalance (~8.6% positive prevalence) natively, without clinical measurements. To achieve this, we engineered five clinically motivated composite features and employed LightGBM gain-based selection to isolate the 12 most predictive variables. Within a strictly leakage-free pipeline, imbalance was managed via the `scale_pos_weight` parameter, and an Optuna-tuned decision threshold (0.5917) was established to prioritize recall. LightGBM achieved the strongest discriminative performance among eight evaluated classifiers, yielding an ROC-AUC of 0.8293, a test F1-score of 0.8225, and an MCC of 0.3201. Crucially, train-test performance showed negligible divergence, confirming robust generalization. These findings demonstrate the viability of using gradient-boosted tree models on demographic, societal, and lifestyle data as highly effective, interpretable screening tools for early cardiovascular risk stratification.

1 INTRODUCTION

Clinical cardiovascular risk assessments traditionally rely on physiological measurements such as electrocardiograms, lipid panels, and blood pressure readings. While highly accurate, these diagnostic tools necessitate in-person medical visits. This requirement can delay early detection, particularly for individuals whose risk factors stem primarily from socioeconomic or lifestyle conditions. Consequently, there is a distinct need for accessible preliminary screening mechanisms that utilize easily obtainable, self-reported data to accelerate early intervention.

This study investigates whether the likelihood of adverse cardiovascular events, specifically myocardial infarction and hypertension, can be reliably predicted using only demographic, societal, and behavioral variables. A major methodological challenge in modeling this type of epidemiological data is severe class imbalance. The target dataset

exhibits a positive disease prevalence of approximately 8.6%, a skew that degrades the performance of standard machine learning algorithms. To address this, we developed a predictive pipeline that assesses cardiovascular risk directly from survey responses, explicitly avoiding data-altering interventions like undersampling or synthetic data generation.

Our analysis leverages the 2020 Behavioral Risk Factor Surveillance System (BRFSS) dataset, which includes 319,795 respondent records. We evaluated eight distinct classifiers within a strictly leakage-free framework. Feature engineering yielded five clinically motivated composite variables, followed by LightGBM gain-based selection to isolate the 12 most predictive factors. Rather than artificially balancing the training set, the pipeline manages the inherent class imbalance natively via the `scale_pos_weight` hyperparameter. The decision threshold was then optimized using Optuna (yielding 0.5917) to

maximize recall, actively prioritizing the minimization of false negatives for clinical safety.

This work offers three primary contributions. First, we establish a predictive pipeline that natively handles extreme epidemiological class imbalance, bypassing the data loss or synthetic artifacts introduced by traditional resampling techniques. Second, we demonstrate that high discriminative accuracy (achieving an ROC-AUC of 0.8293 via LightGBM) is attainable using exclusively non-clinical features. Finally, we provide an interpretable, deployment-ready framework optimized for clinical sensitivity, equipping healthcare professionals with a proactive tool for early-stage risk stratification.

2 REVIEW OF LITERATURE AND RELATED WORK

Heart disease prediction remains a critical area of health informatics, as cardiovascular risk is shaped not only by biological factors but also by demographic conditions and lifestyle habits like physical activity and smoking. While traditional diagnostic models often rely on invasive clinical indicators, recent advances in artificial intelligence enable more adaptive frameworks that utilize self-reported survey data. For instance, (Sadia Tuly et al., 2025) demonstrated that lifestyle variables from the Behavioral Risk Factor Surveillance System (BRFSS) could power resilient stacking ensemble models, identifying these factors as "vital predictors" for cardiac risk. Similarly, (Mageswari et al., 2024) utilized data-driven methods to decode patterns across different age groups and genders, underscoring the importance of demographic stratification. Together, these studies emphasize the effectiveness of non-clinical data in providing scalable, early-warning systems for heart-related issues.

The selection of appropriate algorithms remains a central theme in recent literature, with a growing preference for gradient-boosting frameworks over traditional classifiers. (Rao et al., 2025) introduced a "LightBoost" (LightGBM) model optimized via a teaching-learning seagull algorithm, proving that high-dimensional health data can be processed with high precision. This is complemented by the work of (Deng et al., 2025), who developed an interpretable LightGBM framework that achieved over 90% accuracy on BRFSS data by leveraging SHAP analysis to highlight influential features. While some

research, such as that by (Jansi Rani et al., 2022), focuses on hardware-centric solutions like wearable ECG sensors for real-time monitoring, the prevailing trend for population-level screening favors the computational efficiency and interpretability of ensemble methods like LightGBM.

Further research highlights that effective prediction depends on the model's ability to navigate the complexities of real-world data, specifically geographic and social determinants. (Cetin, 2025) validated that health-related behaviors and conditions in specific populations, such as older adults in the 2020 BRFSS dataset, provide sufficient signal for identifying individuals at risk without clinical measurements. However, these datasets often present challenges like severe class imbalance. (Yanes et al., 2024) tackled this by prioritizing balanced accuracy and macro-recall to ensure that chronic disease models do not overlook high-risk patients. (Wanyonyi et al., 2025) further demonstrated that hybrid ensemble approaches can significantly improve sensitivity—often reaching approximately 90%—when specifically tuned to manage skewed data distributions. These findings suggest that performance metrics must be aligned with clinical safety to be useful in a medical context.

Collectively, these studies underscore that algorithm choice, feature engineering, and robust handling of class imbalance are critical to building reliable heart disease prediction models. Research by (Vu et al., 2025) suggests that early intervention is predicated on a model's ability to minimize false negatives, a priority that shifts the focus from simple accuracy to high recall. Inspired by these developments, our project utilizes LightGBM on the 2020 BRFSS dataset, employing custom feature engineering and an Optuna-tuned decision threshold of 0.5917. By prioritizing recall over accuracy, we aim to provide a more effective decision-support tool for healthcare professionals, ensuring that high-risk individuals are identified for timely intervention.

3 MATERIALS AND METHODOLOGY

3.1 Problem Foundation

The goal of this project is to accurately predict the heart disease status of an individual based on demographic, societal, and lifestyle attributes. The learning objective is to approximate a function that maps input features to a predicted class label (i.e. heart disease), such that prediction error is minimized. The central challenge of this problem is severe class imbalance. In our dataset, positive cases constitute approximately 8.6% of the samples, yielding a ratio of nearly 10.7:1. Under this distribution, a trivial majority class classifier achieves high accuracy while providing no clinical utility. Accordingly, the evaluation criteria and decision threshold must be selected to penalize false negatives more heavily than false positives, reflecting the asymmetric cost inherent to medical screening: a missed diagnosis (false negative) carries substantially greater risk than an unnecessary follow-up referral (false positive).

3.2 Proposed Methodology

The proposed pipeline consists of six sequential stages: data preprocessing, feature engineering, feature selection, data transformation, model training, and threshold optimization.

3.2.1 Stage 1: Data Preprocessing.

All binary survey responses encoded as “yes”/”no” strings are mapped to 1 and 0 respectively. This applies to the attributes HeartDisease, Smoking, AlcoholDrinking, Stroke, DiffWalking, PhysicalActivity, Asthma, KidneyDisease, and SkinCancer. The Sex attribute is encoded as Female = 0 and Male = 1. Nominal multi-level attributes (AgeCategory, GenHealth, Diabetic, and Race) are all retained as categorical typed columns, enabling native handling by gradient-boosted tree frameworks without ordinal encoding. The dataset contains no missing values, so imputation is unnecessary.

3.2.2 Stage 2: Feature Engineering.

Five composite features are constructed from domain knowledge to encode clinically meaningful interactions that may not be directly captured by the

raw attributes. An auxiliary numeric variable, Age_Numeric, is first derived by assigning the midpoint value to each age-category bin. The five engineered features are formally defined in Table 1.

3.2.3 Stage 3: Feature Selection.

A LightGBM surrogate model is fitted on the training partition to derive split-gain feature importances. We reduce our 23-dimensional augmented feature space to a compact subset consisting of the 12 most important features. This selection is performed exclusively on the training data set to prevent information from the held-out partition influencing feature inclusion. Our model should generalize from training data, not memorize any test data.

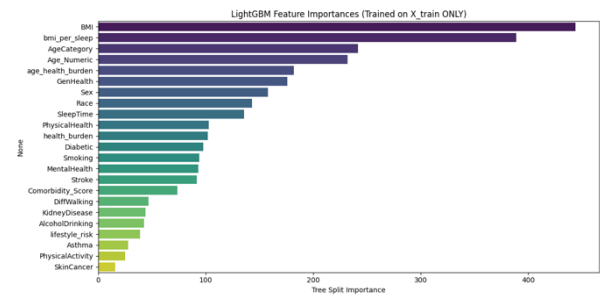


Figure 1: LightGBM split-gain feature importances ranked by contribution, fitted on the training partition only.

3.2.4 Stage 4: Data Transformation.

Tree-based models are invariant to monotonic feature rescaling and therefore require no feature normalization. For models sensitive to feature magnitude (Logistic Regression, K-Nearest Neighbors, Gaussian Naïve Bayes, MLP) Standard Scaling is applied to the selected feature matrix, transforming each feature to zero mean and unit variance. The scaler is fitted on the training set only and subsequently applied to the test set to prevent data leakage.

3.2.5 Stage 5: Model Training.

The primary model, LightGBM, is trained with a positive-class scaling weight of 10.7, directly amplifying the minority class samples proportionally. This approach had better results than SMOTE or majority undersampling in our experiments. A 5-fold stratified cross-validation is conducted on the training set. The final model is subsequently fitted on the full training set. LightGBM was selected after evaluating eight candidate algorithms (KNN, Logistic Regression, Random Forest, XGBoost, Naive Bayes, MLP, LightGBM, and CatBoost), as it delivered the

strongest and most consistent performance while natively handling class imbalance via the `scale_pos_weight` hyperparameter.

3.2.6 Stage 6: Threshold Optimization.

The default decision threshold of 0.5 is replaced by an optimal threshold identified using the Optuna framework over 100 trials. The threshold selected for final reporting (0.5917) is tuned to favor recall over precision, consistent with our objective of minimizing missed diagnoses.

3.3 Proposed Architecture

The overall system is organized as a linear ML pipeline with strict separation between Training and Testing data. The pipeline proceeds through the following stages in order: Raw Survey Data → Data Preprocessing → Feature Engineering → Stratified Train/Test Split → Feature Selection (on training set only) → Feature scaling (training set only, for applicable models only) → LightGBM Training → Optuna optimization → Out-of-sample Evaluation → Model serialization (for direct comparison). All parameter learning sets are fitted exclusively on the training partition and applied to the test partition without refitting. This design ensures that no information from the test set influences any stage of model construction. The trained artefacts are serialized for reproducibility on new patient records.

3.4 Description of Dataset

The dataset used in this work is the 2020 Behavioral Risk Factor Surveillance System survey, as cleaned and made available by Kamil Pytlak. The BRFSS is an annual CDC-administered telephone survey collecting self-reported health risk data from U.S. residents across all 50 states. The cleaned version employed herein comprises 319,795 records and 18 features. The binary target variable, `HeartDisease`, has a positive-class prevalence of approximately 8.6%, corresponding to a 10.7:1 negative-to-positive ratio. Features span three domains: demographic attributes (`Sex`, `AgeCategory`, `Race`), health status indicators (`GenHealth`, `Diabetic`, `Stroke`, `Asthma`, `KidneyDisease`, `SkinCancer`), and lifestyle/behavioral measures (`BMI`, `Smoking`, `AlcoholDrinking`, `PhysicalActivity`, `PhysicalHealth`, `MentalHealth`, `SleepTime`). The dataset is partitioned into 255,836 training samples and 63,959 test samples via stratified

sampling, preserving the natural class distribution in both datasets.

3.5 Hyperparameter Configuration

Table 1: Engineered Feature Definitions.

Feature Name	Formula	Rationale
Comorbidity_Score	Stroke + Asthma + KidneyDisease + SkinCancer	Aggregate comorbidity burden
health_burden	PhysicalHealth + MentalHealth	Total self-reported poor-health days
bmi_per_sleep	BMI / (SleepTime)	Metabolic load relative to recovery time
age_health_burden	Age_Numeric * health_burden	Age-amplified illness severity interaction
lifestyle_risk	Smoking+(1-PhysicalActivity)	Composite lifestyle risk index

Table 2: Primary Model (LightGBM) Hyperparameters

Hyper-parameter	Value	Description
n_estimators	500	Number of boosting rounds
max_depth	5	Max depth per tree
learning_rate	0.0108	Gradient step size
num_leaves	104	Max leaf nodes per tree
subsample	0.7717	Row subsampling per tree
colsample_bytree	0.5940	Feature subsampling ratio per tree
min_child_samples	50	Minimum samples required per leaf
scale_pos_weight	10.7	Positive-class loss scaling weight
random_state	42	Reproducibility seed

Decision threshold	0.5917	Optuna-optimized over 100 trials
--------------------	--------	----------------------------------

3.6 Objective Functions

3.6.1 Loss Function

LightGBM internally uses Binary Cross-Entropy as its cost function. The loss is computed internally by the LightGBM framework during gradient optimization and is not directly reported as an evaluation metric.

Table 3: Performance Evaluation Metrics

Metric	Formula	Purpose
Accuracy	$(TP + TN) / N$	Overall correctness. Deemphasized due to class imbalance
Precision	$TP / (TP + FP)$	Fraction of positive predictions that are true positives
Recall	$TP / (TP + FN)$	Fraction of true positive cases correctly identified. Best clinical measure
F1-Score	$2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$	Harmonic mean of precision and recall
ROC-AUC	Area under the Receiver Operating Characteristic curve	Threshold-independent ranking ability
MCC	$(TP \times TN - FP \times FN) / \sqrt{((TP+FP)(TP+FN)(TN+FP)(TN+FN))}$	Balanced measure accounting for all four confusion matrix cells

3.7 Benchmarks

Table 4: Comparative Baseline Models

Model	Type	Rationale
LightGBM (primary model)	Gradient Boosting	Handles categorical features natively, optimized for imbalanced tabular data
Logistic Regression	Linear classifier	Standard probabilistic baseline; interpretable decision boundary
Random Forest	Ensemble (bagging)	Strong tree baseline; contrasts boosting versus bagging strategies
XGBoost	Gradient boosting	Widely-used boosting benchmark for tabular data
CatBoost	Gradient boosting	Native categorical feature handling; direct competitor to LightGBM

All baseline models are trained on identical training data and evaluated on identical testing data, ensuring a fair and controlled comparison. Class imbalance within each model through the mechanism most appropriate for its implementation: `scale_pos_weight` for LightGBM and XGBoost, `class_weight = balanced` for LogisticRegression and RandomForest, and `auto_class_weights = balanced` for CatBoost.

4 EXPERIMENT AND RESULTS

4.1 Experimental Setup

All experiments were conducted in a Google Colab notebook, Python3 using scikit-learn, LightGBM, XGBoost, and Catboost frameworks, and the pandas and numpy libraries. The dataset was partitioned into a training set of 255,836 samples and a test set of 63,959 samples using stratified random sampling with a fixed seed of 42, preserving the 8.6% positive-class prevalence across both partitions. The primary model was evaluated using a 5-fold stratified cross validation on the training partition, followed by a final evaluation on the test partition. All comparison models were trained and evaluated on the identical partitions to ensure a controlled and fair comparison. The decision threshold for the primary model was tuned via Bayesian optimization using Optuna over 100 trials, yielding a final threshold of 0.5917.

4.2 Primary Model Performance

Table 5 reports the performance of the proposed LightGBM model across all evaluation metrics for both the training and test data. The close agreement between training and test results across all metrics indicate that the model generalizes well into unseen data with no substantial overfitting.

Table 5: LightGBM Primary Model- Training and Test Performance

Metric	Training	Testing
Accuracy	0.7197	0.7784
Precision	0.9093	0.9048
Recall	0.7197	0.7784
F1	0.7806	0.8225
ROC-AUC	0.8321	0.8293
MCC	0.3071	0.3201

The model achieves an ROC-AUC of 0.8293, indicating a strong discriminative ability across all possible decision thresholds. Precision of 0.9048 reflects a low false positive rate among predicted cases, while recall of 0.7784 captures approximately 78% of true positive cases. The MCC of 0.3201 confirms meaningful predictive signal beyond what a majority-class baseline would achieve.

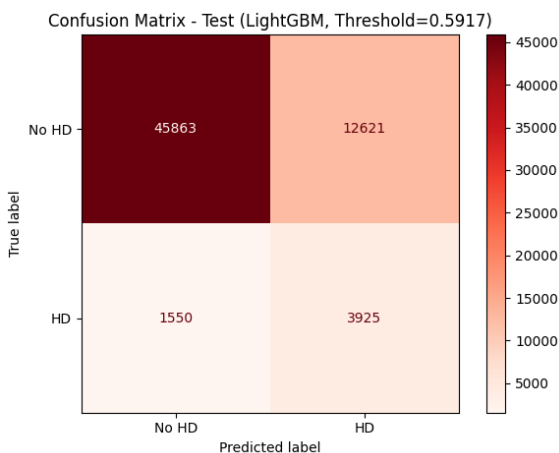


Figure 2: Confusion matrix for the LightGBM model on the held-out test set (threshold = 0.5917).

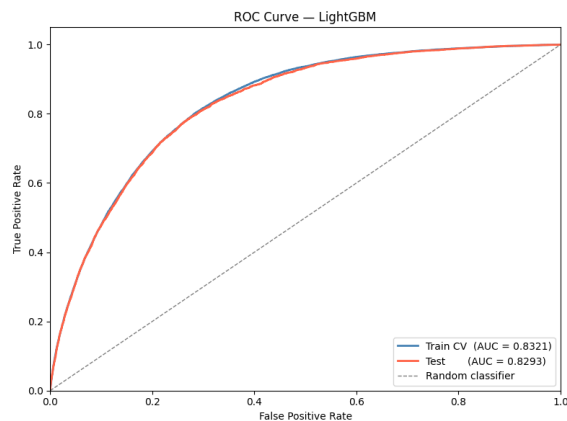


Figure 3: Receiver operating characteristic curves comparing training (5-fold CV) and test set performance for the LightGBM model.

4.3 Comparative Model Evaluation

Table 6 presents the test set performance of all models evaluated under identical experimental conditions.

Table 6: Comparative Model Results

Model	F1	AUC	MCC
LightGBM	0.7789	0.8293	0.3040
Logistic Regression	0.7901	0.8234	0.3051
Random Forest	0.7812	0.8250	0.3018
XGBoost	0.7842	0.8281	0.3039
CatBoost	0.7801	0.8285	0.3038

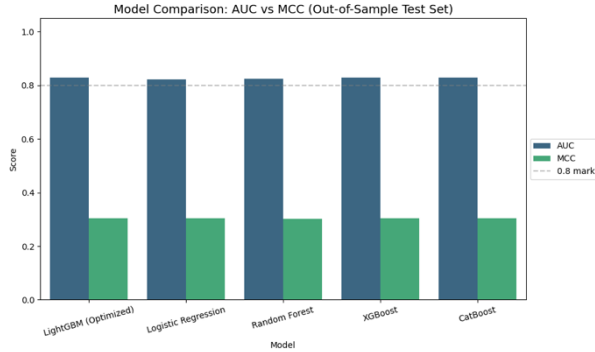


Figure 4: Comparative AUC and MCC scores across all evaluated models on the held-out test set.

Several observations merit discussion. While Logistic Regression records marginally higher F1-Score and MCC, LightGBM achieves the highest ROC-AUC across all evaluated models. In the context of a medical screening tool targeting a severely imbalanced population, ROC-AUC is the most appropriate comparative criterion. A model with superior AUC retains greater flexibility for threshold adjustment in deployment contexts where the clinical tradeoff between false positives and false negatives may shift. LightGBM is therefore retained as our proposed model. The performance difference between gradient-boosted models (LightGBM, XGBoost, CatBoost) are narrow, spanning less than 0.001 AUC, confirming the robustness of this type of model for this task.

5 DISCUSSION

5.1 Analysis of Results

The strong performance of the proposed LightGBM model can be attributed to three key design decisions. First, class imbalance was handled natively via the `scale_pos_weight` parameter (set to 10.7), which directly scales the loss contribution of positive-class samples during training. This approach outperformed both SMOTE and majority undersampling in our experiments, as it avoids introducing synthetic artifacts or discarding valuable majority-class information. Second, Optuna-driven Bayesian optimization of hyperparameters (`learning_rate` = 0.0108, `num_leaves` = 104) enabled fine-grained control over model complexity, reducing overfitting while maintaining discriminative power. Third, the optimized decision threshold of 0.5917 shifted the precision-recall tradeoff toward higher recall, directly minimizing missed diagnoses (false negatives),

which is the more clinically costly error in a screening context.

The five engineered features, particularly `age_health_burden` and `Comorbidity_Score`, contributed meaningful predictive signal by encoding domain-relevant interactions absent from the raw survey attributes. The ROC-AUC of 0.8293 indicates that the model correctly ranks a randomly selected positive case above a randomly selected negative case approximately 83% of the time. Precision of 0.9048 confirms that the vast majority of positive predictions correspond to true cases, while recall of 0.7784 indicates the model identifies roughly 78% of actual heart disease cases. The MCC of 0.3201, though moderate in absolute terms, confirms that the model produces predictions meaningfully above chance given the severe 10.7:1 class imbalance, where a majority-class classifier would achieve an MCC of zero.

The comparative evaluation further validates the choice of LightGBM. Although Logistic Regression achieved a marginally higher F1-Score, LightGBM attained the highest ROC-AUC among all evaluated models, making it the most flexible option for deployment scenarios where the clinical cost tradeoff between sensitivity and specificity may vary. The narrow performance spread among all three gradient-boosted models (less than 0.001 AUC) also confirms the general robustness of tree-based ensembles for this task.

These results demonstrate that cardiovascular risk can be predicted reliably using only self-reported survey data, without requiring clinical measurements. This positions the model as a viable low-cost screening tool, particularly for underserved populations with limited access to healthcare, where early flagging of at-risk individuals could facilitate timely intervention and improved outcomes.

5.2 Limitations

Several limitations of this work should be acknowledged. First, the BRFSS dataset relies entirely on self-reported survey responses, which are subject to recall bias, social desirability bias, and measurement error. Respondents may underreport behaviors such as smoking or alcohol consumption, or inaccurately assess their own general health status, potentially attenuating the true predictive signal of these features.

Second, the model was developed and evaluated on a single cross-sectional dataset from the year 2020.

No temporal validation was performed across different survey years, and no external validation on an independent population was conducted. Consequently, the generalizability of the reported performance to other demographic distributions or time periods remains unconfirmed.

Third, the binary target variable (HeartDisease: Yes/No) collapses a heterogeneous spectrum of cardiovascular conditions into a single label. This simplification may obscure clinically meaningful distinctions between conditions such as myocardial infarction, coronary artery disease, and hypertensive heart disease, each of which may have different risk profiles.

Finally, while the model achieves strong ranking performance (AUC = 0.8293), the moderate MCC of 0.3201 reflects the inherent difficulty of producing well-calibrated binary predictions under extreme class imbalance. In a real-world deployment, the model would function most effectively as a preliminary screening tool to flag individuals for further clinical evaluation, rather than as a standalone diagnostic instrument.

6 CONCLUSIONS

In this project, we developed a model to predict heart disease risk using demographic, societal, and lifestyle data. Our results show that even without clinical measurements, it is still possible to identify meaningful patterns related to cardiovascular risk. Out of all the models we tested, LightGBM performed the best, achieving an ROC-AUC of about 0.83 and showing consistent performance between training and testing data. A key part of our approach was handling class imbalance directly within the model using the `scale_pos_weight` parameter, instead of relying on techniques like SMOTE. We also found that engineered features such as comorbidity score, health burden, and lifestyle risk helped improve the model's predictions.

6.1 Future Work

Overall, this work suggests that self-reported data can be useful for early heart disease screening, especially in situations where access to medical testing is limited. For future work, the model could be improved by testing on additional datasets, further tuning model parameters, and exploring how it performs in a real-world healthcare setting.

ACKNOWLEDGMENTS

The authors would like to thank Dr. Bonaventure Chidube Molokwu for his guidance and instruction throughout this project, as well as the Department of Computer Science at California State University, Sacramento for providing the resources and environment necessary to conduct this work.

REFERENCES

- Cetin, Z. (2025). Predicting Heart Disease Using Machine Learning Models. Williams Honors College, Honors Research Projects, 2005. The University of Akron.
- Deng, L., Lu, K., & Hu, H. (2025). An interpretable LightGBM model for predicting coronary heart disease: Enhancing clinical decision-making with machine learning. *PLOS One*, 20(9), e0330377.
- Jansi Rani, S. V., Sarath Chandran, K. R., Ranganathan, A., Chandrasekharan, M., Janani, B., & Deepsheka, G. (2022). Smart wearable model for predicting heart disease using machine learning. *Journal of Ambient Intelligence and Humanized Computing*, 13, 4321-4332.
- Mageswari, P. U., Annakumari, R., Kavitha, S., Sah, S., & Kumar, G. (2024). Decoding Heart Health: A Predictive Model for Demographic-Based Cardiovascular Risk. *Proceedings of the 2024 4th International Conference on Technological Advancements in Computational Sciences (ICTACS)*. IEEE.
- Rao, C., Lian, J., Wen, J., & Chen, L. (2025). Predicting risk of coronary heart disease using a teaching-learning seagull optimization algorithm with the lightboost based gradient boosting machine. *International Journal of Machine Learning and Cybernetics*, 16, 8195-8214.
- Tuly, S., Majumder, S., Chowdhury, A., Habib, A., & Suha, S. A. (2024). Predicting Heart Disease Risk with Lifestyle Factors: Uncovering Vital Predictors via Feature Selection and Machine Learning Techniques. In *3rd International Conference on Computing Advancements (ICCA 2024)*. ACM.
- Vu, T., Kokubo, Y., Inoue, M., Yamamoto, M., Mohsen, A., Martin-Morales, A., et al. (2025). Machine Learning Model for Predicting Coronary Heart Disease Risk: Development and Validation Using Insights From a Japanese Population-Based Study. *JMIR Cardio*, 9, e68066.
- Wanyonyi, M., Morris, Z. N., Musyoka, F. M., & Kitavi, D. M. (2025). Enhanced machine learning and hybrid ensemble approaches for Coronary Heart Disease prediction. *PLOS One*, 20(12), e0328338.
- Yanes, N., Jamel, L., Alabdullah, B., Ezz, M., Mostafa, A. M., & Shabana, H. (2024). Using Machine Learning for Detection and Prediction of Chronic Diseases. *IEEE Access*, 12, 177674-177691.